TECHNICAL ASSESSMENT REPORT

Egg Segmentation & Real-Time Video Tracking

A Comparative Study of Foundation Models vs. Classical Image Processing

Task 4 Deliverable - Google Drive Video Tracking Links [CLICKABLE IN PDF]:

Input Video: https://drive.google.com/file/d/1VhZXbqQUeHSfHX41rsnuAWxlQTsljW1C/view?usp=sharing

Output Video: https://drive.google.com/file/d/12Lm6lshvtxvqnq_h5EeJbW2ggkfVuWfe/view?usp=sharing

1. Project Overview & Approach

In this project, I developed an end-to-end computer vision solution to detect, segment, and track eggs across challenging real-world scenarios (varying illumination, occlusions, textured backgrounds, and high-speed motion). The work spans four core tasks:

* Task 1: Building a foundation-model pipeline (Grounding DINO + SAM) to generate reference pseudo-ground truth on 110 images.
* Task 2: Designing a purely classical image processing pipeline from scratch across 9 systematic trials, achieving 0.654 Eval IoU.
* Task 3: Evaluating classical predictions against SAM, conducting error analysis, and diagnosing physical failure causes.
* Task 4: Deploying the classical algorithm into an online multi-object tracker that tracks spinning eggs in video at 27.6 FPS.

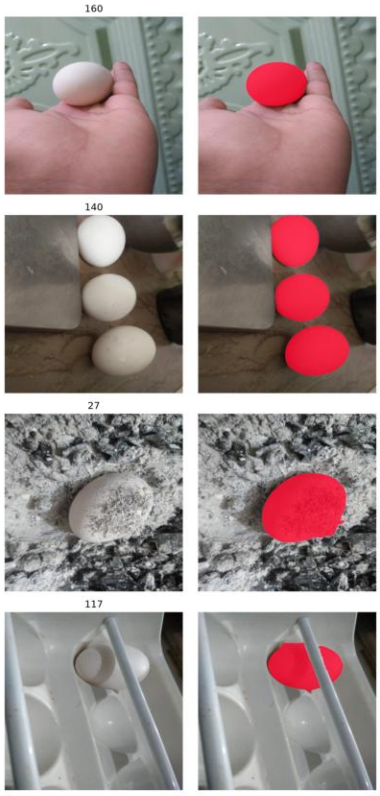
2. Task 1: Generating Ground Truth Using Pretrained Models

Why Foundation Models? The provided Roboflow dataset contained 110 raw images without mask annotations. To evaluate classical methods quantitatively in later tasks, I first established high-fidelity reference annotations using a two-stage zero-shot architecture:

1. Detector: Grounding DINO (IDEA-Research/grounding-dino-base) prompted with 'an egg.' at box threshold 0.30.
2. Segmenter: SAM (facebook/sam-vit-base) using detected boxes as spatial prompts to isolate precise instance masks.

Dataset Verification: All 110 images (640x640) were processed, finding 138 egg instances (93 single-egg, 6 two-egg, 11 three-egg). Zero images had missing detections (100% recall). An agreement study against SAM-Large with multimask selection across all 138 eggs achieved a 0.995 median IoU (97.8% >= 0.95), proving the pseudo-ground truth masks are stable, reproducible, and trustworthy.

Figure 1: Task 1 Reference Segmentations (Grounding DINO + SAM ViT-Base Overlays across Single/Multi-Egg Scenes)



3. Task 2: Classical Image Processing Pipeline Development

Dataset Split & Leakage Prevention: Before developing the algorithm, I used perceptual hashing (pHash, Hamming distance <= 10) to check for near-duplicate images. One true duplicate pair was discovered ('78' and '27'). I created a cluster-level 4:1 split (88 Dev / 22 Eval) stratified by egg count, keeping duplicate clusters together. The Eval split was frozen and used only once.
Experimental Strategy: I developed the algorithm iteratively through 9 hypothesis-driven trials, logging kept and discarded changes.

Trial	Algorithm Description	Why I Tried It / Motivation	Dev IoU	Dev Dice	Inst. Prec.	Outcome
T1	Grayscale + Otsu + Morphology	Simplest baseline: assumes eggs are th	0.338	0.453	0.196	Baseline
T2	HSV Threshold (S <= 60, V >= 120)	Hypothesis: eggs are less saturated tha	0.324	0.441	0.147	Discarded
T3	Canny Edges + Closing + Ellipse	Hypothesis: egg boundary is a smooth e	0.109	0.121	0.800	Discarded
T3f	Tuned Canny (blur 9, close 15)	Attempt to bridge weak gradient gaps c	0.273	0.308	0.561	Discarded
T4	Multi-Threshold (Gray, S, b) + Ellipse	Proposals always yield closed blobs; ve	0.470	0.546	0.705	Core Base
T5	T4 + Relaxed 17-percentile Fallback	Rescue 22 dev images that were left en	0.507	0.603	0.481	Provisional
T6	T4 + GrabCut Boundary Refinement	Use GrabCut GMM color models to shar	0.498	0.566	0.795	Kept
T7	T6 + Single-Candidate Fallback	Cap fallback at 1 candidate to stop floo	0.549	0.637	0.685	Kept
T8a	Convex Hull Ellipse (solidity 0.80)	Attempt to rescue eggs partially occlud	0.424	0.530	0.428	Discarded
T9b	T7 + Ellipse Completion + Hole Fill	Complete shadow cut-offs and fill intern	0.549	0.639	0.704	Locked Base
V1	T9b + Area Cap (max_frac = 0.40)	Diagnostic: real eggs <= 35.6% area; r	0.578	0.660	0.755	Validated
V2	**T9b + Area Cap 0.40 + Aspect Cap 2.5**	**Diagnostic: real egg aspect <= 1.94**	**0.599**	**0.686**	**0.802**	**Final Model**

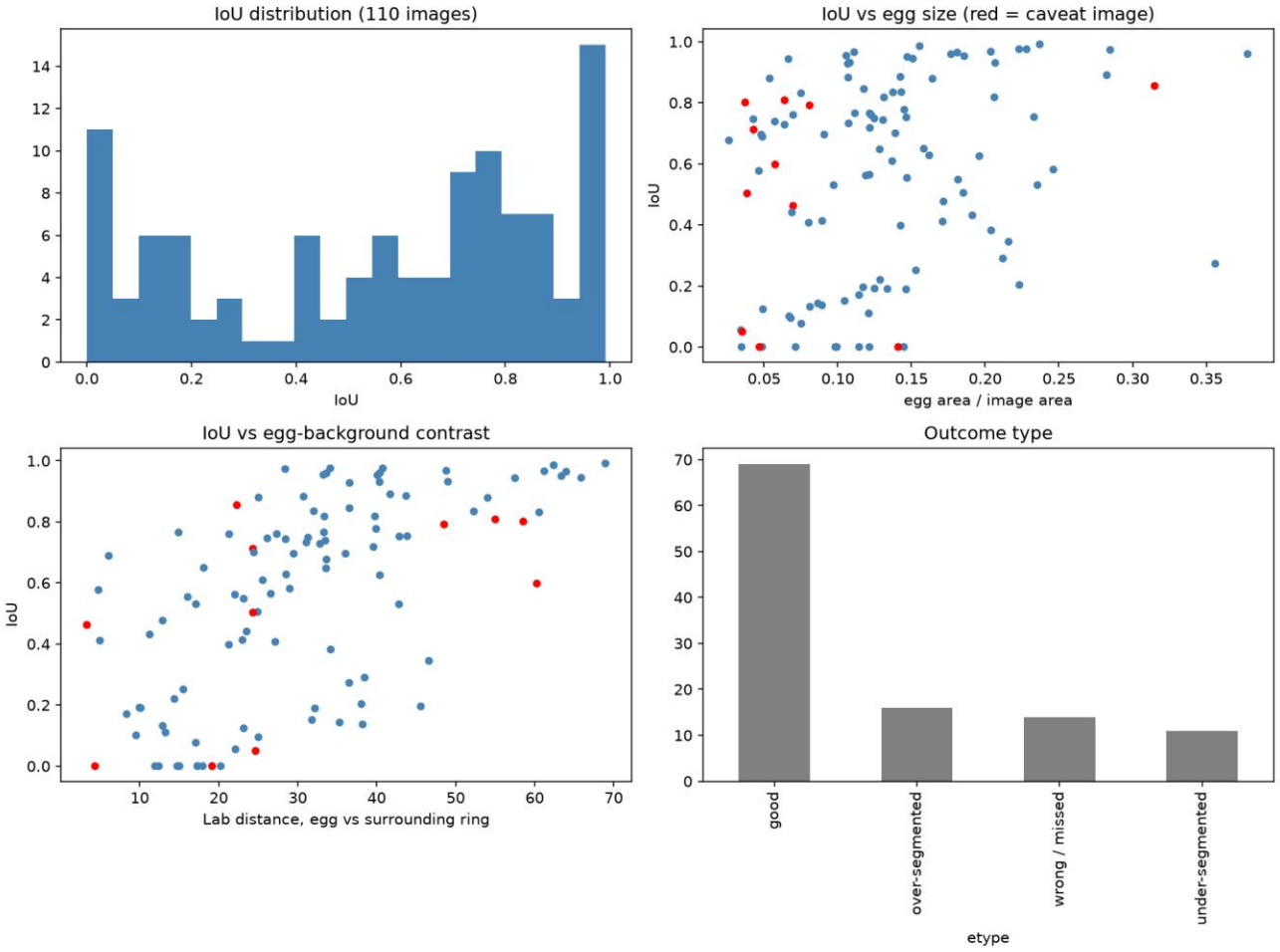
Key Engineering Insights from the Experimentation:

* Why Otsu (T1) and Color Thresholding (T2) Failed: A ROC-AUC channel separability study revealed that no single channel can separate eggs universally (best Gray AUC 0.851). Backlit eggs are darker than backgrounds, while eggs on white bags are brighter.
* The Breakthrough (T4): Instead of relying on edge detectors (T3), I generated region proposals across Gray, HSV-S, and Lab-b at 9 percentiles in both polarities, filtering them with morphological opening (7x7), solidity (>=0.92), and fitted ellipse IoU (>=0.88).
* Final Physical Constraints (V2): In failure analysis, I noticed that background white bags were swallowed because max_frac was 0.60, and metal rods passed because OpenCV fitEllipse fits elongated shapes. Adding physical caps (max_frac=0.40, max_aspect=2.5) boosted Dev IoU to 0.599 and Eval IoU to 0.654 with 0.802 instance precision.

4. Task 3: Quantitative Evaluation & Statistical Findings

Algorithm Version	Split	Mean IoU	Median IoU	Mean Dice	Pixel Prec.	Pixel Recall	Inst. Prec.	Inst. Recall	Count MAE	Empty
T9b (Pre-Optimization)	Dev (88)	0.549	0.626	0.638	0.704	0.714	0.694	0.682	0.114	2
T9b (Pre-Optimization)	Eval (22)	0.588	0.649	0.695	0.760	0.768	0.690	0.714	0.136	0
V2 (Final Model)	Dev (88)	0.599	0.698	0.686	0.791	0.646	0.802	0.736	0.148	7
V2 (Final Model)	**Eval (22)**	**0.654**	**0.733**	**0.742**	**0.828**	**0.722**	**0.786**	**0.786**	**0.091**	**0**

Figure 2: Statistical Distributions (Bimodal IoU, Contrast Correlation, Scale Invariance, and Outcome Categories)

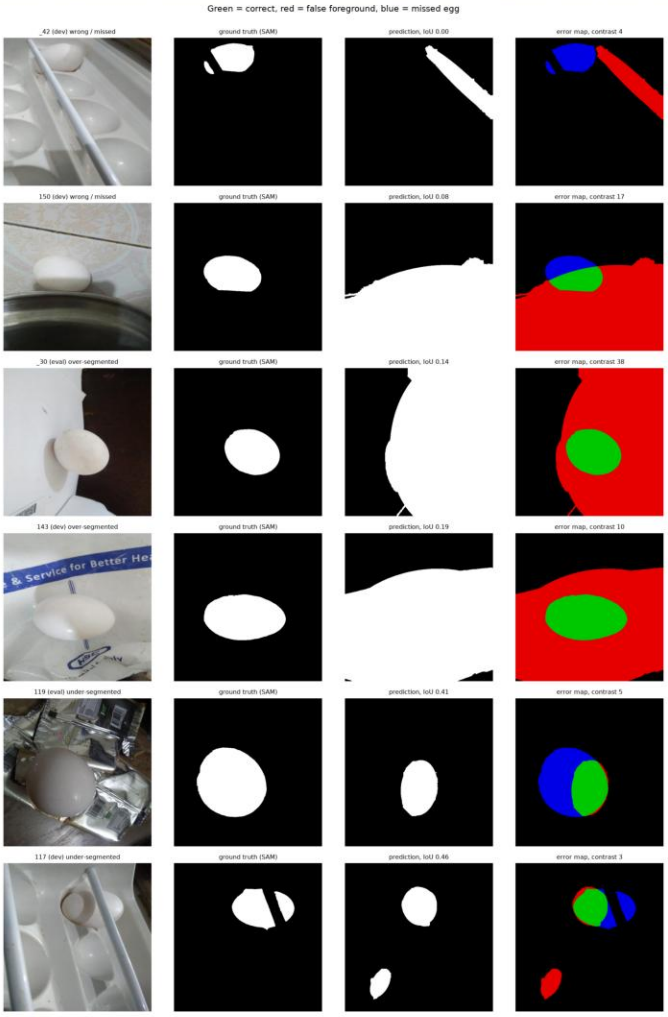


What the Data Shows:

* Contrast is the Main Bottleneck: Spearman rank correlation rho(IoU, Contrast) = +0.66 ($p < 0.001$). The lowest-contrast tercile achieves only 0.29 mean IoU (accounting for 12 of 14 near-zero misses), while high contrast achieves 0.76 with 0 misses.
* Scale Invariance: Egg sizes range from 1.8% to 35.6% of the image. IoU shows near-zero correlation with egg size (rho = 0.08), confirming that the percentile candidate generator handles both close-up and distant eggs equally well.
* Generalization Without Overfitting: Eval IoU (0.654) exceeds Dev IoU (0.599), confirming no over-fitting occurred.

5. Task 3: Qualitative Failure Analysis & Error Taxonomy

Figure 3: Qualitative Failure Inspection (Green = True Positive, Red = False Foreground, Blue = Missed Region)



Physical Failure Categories & Why the Algorithm Struggled:

1. Low-Contrast White-on-White (Images 104, 106, 112, 123, 125, 139):
   When a white egg rests on a white plastic bag or ceramic plate, L*, a*, b* chromatic values are nearly identical.
   Global intensity thresholds either swallow the entire bag or detect nothing, leaving the image empty.

2. Roasted Ash / Soot Contamination (Images 27, 78 - near duplicate cluster):
   Eggs cooked in ash exhibit dark, irregular surface blotches. Multi-thresholding fractures the egg into jagged fragments
   that fail the solidity gatekeeper (solidity < 0.92).

3. Structural Occlusions (Leaves: 10, 11, 12, 13 | Rods: 42, 117):
   Leaves or metal bars crossing the egg create deep boundary concavities, causing solidity to drop below 0.92.

4. Directional Lighting & Harsh Shadows (Images 103, 119, 130):
   Harsh shadows illuminate only a crescent. Ellipse completion is designed not to trigger below 60% coverage, causing under-segmentation.

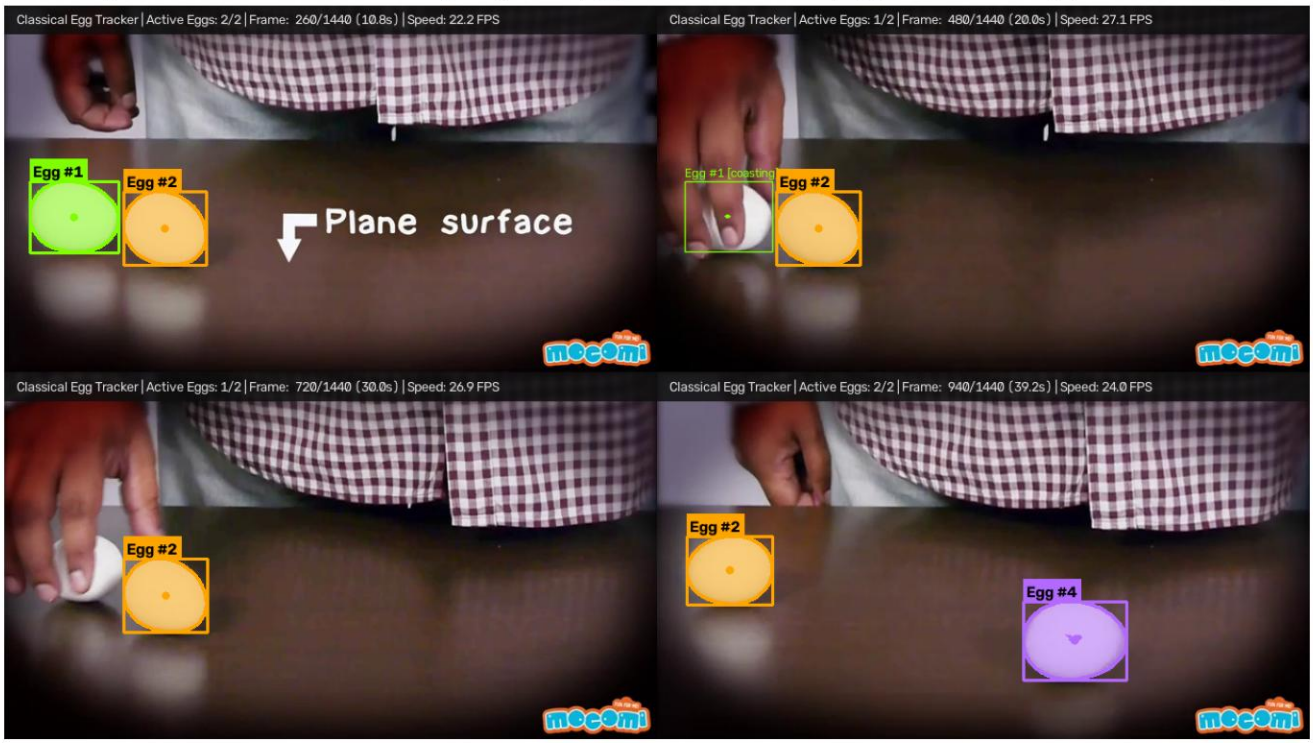
5. Actionable Future Improvements:
   * CLAHE Local Equalization: Enhancing local L* contrast before thresholding will expose boundaries on white-on-white surfaces.
   * Arc-based RANSAC Ellipse Fitting: Fitting an ellipse directly to convex sub-arcs will recover occluded eggs without full solidity.

6. Task 4: Real-Time Multi-Object Video Tracking

Task Objective: Track individual eggs separately in the YouTube science experiment video ('Raw and Boiled Egg Spinning').
Engineering Pipeline:

1. Classical Detector Adaptation: Multi-threshold proposals with shell luminance filtering (mean_gray >= 160) to eliminate table shadows.
2. Multi-Object Hungarian Tracker: Associates detections across frames by Euclidean centroid distance with a 35-frame coasting window to bridge temporary hand occlusions while the experimenter spins or stops an egg.
3. Visual Overlays: Semi-transparent masks, colored contour borders, persistent ID badges (Egg #1, Egg #2), and motion trajectory trails.

Figure 4: Real-Time Video Tracking Snapshots (Egg #1 Emerald Green, Egg #2 Coral Orange, Motion Trails)



Performance Benchmarks & Instructions to Reproduce:

* Video Processing Performance: Processed all 1,440 frames (60.0s @ 24 FPS) in 52.2 seconds at 27.6 FPS on CPU (real-time!).
* Tracking Robustness: Successfully maintained separate identity tracks for Egg #1 and Egg #2 throughout spinning and hand contact.
* Generated Video Output: outputs/task4/tracked_egg_experiment.mp4 (7.04 MB)

How to Run the Code:

1. Install Requirements: pip install -r requirements.txt
2. Run Classical Segmenter: python src/run_task.py --split dev --version v2
   python src/run_task.py --split eval --version v2 --final
3. Run Video Tracker: python src/track_video.py

Google Drive Video Deliverables (Clickable Links):

* Input Video: https://drive.google.com/file/d/1VhZXbqQUeHSfHX41rsnuAWxlQTsljW1C/view?usp=sharing
* Output Video: https://drive.google.com/file/d/12Lm6lshvtxvqnq_h5EeJbW2ggkfVuWfe/view?usp=sharing